Fewer empty shelves, before the doors open

Larkwell Foods · on-shelf availability, from raw data to the shop floor

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Every morning, £1.09m of sales Larkwell can fix is at risk on 127,000 shelves

What this delivers

- £23.7m a year of sales recovered in the base case (range £15.8m to £39.5m)
- 328,000 store hours a year given back
- 98% of empty shelves explained automatically, every morning
- From morning snapshot to action at the shelf in minutes
- Runs for roughly £40k to £90k a year

Why shelves are empty today

Root cause	Gaps and lows	Sales at risk a day	Who can act
Sold more than forecast	30,989	£525k	Ordering team
Delivery arrived short	40,711	£232k	Depot and buying
Order blocked or cancelled	24,491	£143k	Depot
Phantom stock	16,297	£97k	Store
Written off as waste	10,437	£65k	Store and buying
Market-wide shortage	16,573	£90k	Substitute
Planned range exit	62,251	£372k	No action needed
Not yet known	4,085	£24k	Analyst

What changes, for each person

Store colleague

Scans the shelf

Gets the cause and the next action in milliseconds

Logs what they did in one tap

Head of Availability

One governed view

98% of gaps attributed

Supplier and depot evidence

Sees every action

COO

Fixable lost sales falling

Store time given back

One number everyone trusts

Value for the COO

Lever	Conservative	Base	Stretch
Sales recovered a year	£15.8m	£23.7m	£39.5m
Store time given back a year	£4.3m	£4.3m	£4.3m
Run cost a year (indicative)	up to £90k	up to £90k	up to £90k
Return on run cost	over 170x	over 260x	over 430x

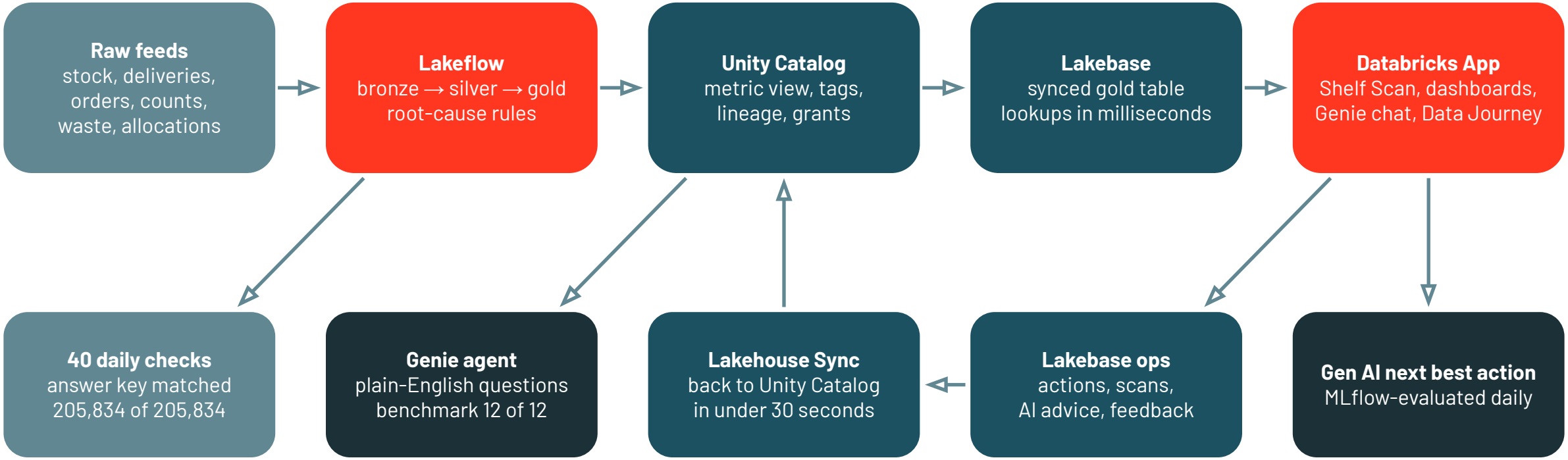
Value for the Head of Availability

KPI	Today	With this build
Root cause confirmed	Manual, the next day	98%, every morning
Time to action	The next day's report	Minutes
Supplier accountability	Anecdotal	Oldcastle Confectionery: 59% fill rate against a 99.0% contract
Depot accountability	Anecdotal	Merseyvale DC: 6,078 blocked lines across 130 stores
Planned exits chased	Often	Never: "no action needed"

What I'll show you

- Head office sees what's wrong, and why, by 6am
- A colleague fixes it at the shelf
- Anyone can ask, and the loop closes
- Then: how it's built, and what it's worth

One connected journey



A Lakeflow Job runs the day: land → pipeline → Lakebase refresh → 40 checks → AI evaluation

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Under the hood

Data and governance

Auto Loader ingests 10 feeds incrementally

38 data-quality expectations

Eight-step root-cause cascade, proven by an answer key: 205,834 of 205,834

Metric view, governed tags (pii), lineage, read-only app

Serving and AI

Lakebase lookups in milliseconds: over 100 times faster than the warehouse

Colleague actions back in Unity Catalog in under 30 seconds

The LLM words the playbook action; it never decides it

Daily MLflow evaluation with calibrated judges; Genie 12 of 12

Proof it works

Check	Result
Gaps and lows matching the answer key	205,834 of 205,834
Daily checks across the journey	40 of 40, every stage, every morning
App end-to-end test	65 of 65
Shelf Scan lookup	2.9 ms on Lakebase against 504 ms on the warehouse
A logged action back in Unity Catalog	14 seconds
Genie benchmark	12 of 12
AI advice evaluation (48 cases)	100% on all eight checks; judges calibrated on 19 hand-labelled answers

Decisions and trade-offs

Decision	Why	Trade-off
Rules cascade, not a model, for root cause	Explainable to a colleague, testable row by row	Machine learning saved for prediction, the next phase
Lakebase for Shelf Scan lookups	Over 100 times faster, built for thousands of users	A governed second copy
The app writes to Lakebase, not Delta	Transactional speed; read-only lakehouse access	Lakehouse Sync is in Beta
The LLM words the advice; it never decides it	No invented numbers or causes	Less "creative" advice, by design
Synthetic data with an answer key	Proves the pipeline on every run	Real-data noise to validate in the pilot

Risks and how we handle them

Risk	How it's handled
Adoption in stores	Fits the handheld routine; actions take one tap; the pilot measures use
Feed data quality	38 expectations; the daily checks fail loudly before anyone sees a number
Security and privacy	Unity Catalog grants; pii tagged; no personal data in Genie's views; a read-only app
Cost	Serverless, autoscaling Lakebase, model-cost controls
Wrong AI advice	Playbook-driven; a daily evaluation with thresholds; a template fallback
Beta feature (Lakehouse Sync)	A federated read over the Lakebase catalog as the fallback

Value in 12 weeks

Weeks	What	What it proves
0-2	Connect real stock, delivery and order feeds with Lakeflow; check the rules against 2 weeks of history	The rules match reality
3-8	Pilot in 20 stores, against 20 control stores	Controllable gap rate, time to action, colleague minutes
9-12	Roll out by division; supplier and depot scorecards	Value at scale

The ask: approve the pilot, with your availability lead, two data engineers, and access to the feeds.

Appendix: Value model assumptions

Input	Value	Source
Forecast demand	£22.6m a day (£8.2bn a year)	Build data
Controllable sales at risk	£1.09m a day	Build data
Share actually lost to the retailer	40%	Gruen, Corsten and Bharadwaj (2002)
Prevented by faster, targeted action	10%, 15% or 25%	Assumption, to test in the pilot
Manual checks per store per day	20, at 3 minutes saved each, £13 an hour	Assumption

Appendix: Run cost (indicative)

Component	Estimate a year
Lakeflow pipeline and daily job	Low thousands of pounds
Lakebase (1-8 CU)	Under £10k
Databricks App	Under £10k
SQL warehouse	£10k-£20k
LLM next best action	£10k-£40k
Total	**About £40k-£90k**

Appendix: Root-cause rules, in order

#	Root cause	Evidence
1	Planned range exit	The line is exiting the store's range
2	Market-wide shortage	The allocation was cut for a market shortage
3	Delivery arrived short	Delivered less than ordered
4	Order blocked or cancelled	Latest order status is blocked or cancelled
5	Phantom stock	A stock count removed stock that wasn't there
6	Waste	Units written off
7	Sold more than forecast	Sales above forecast
8	Not yet known	No evidence: an analyst investigates



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